# AEGIS

Capital preservation research
Quantum-X · Serendib University Endowment
USD 10 million proposed mandate

# The investment decision

ADJUST the mandate before funding
Resolve stress tolerance and the spending objective

# A conservative policy framework

10+ year horizon · provisional 4% nominal objective
AGG 35–70% · gold 5–20% · equities 20–50%
No leverage; daily rebalancing is a research assumption

# Data before decisions

5,470 aligned adjusted closes · seven eligible ETFs
Training through 2018 · development 2019–2022
Test 2023-01-03 to 2026-09-30

# The proposed allocation

Training variance minimization
The AGG cap binds; bond concentration remains a risk



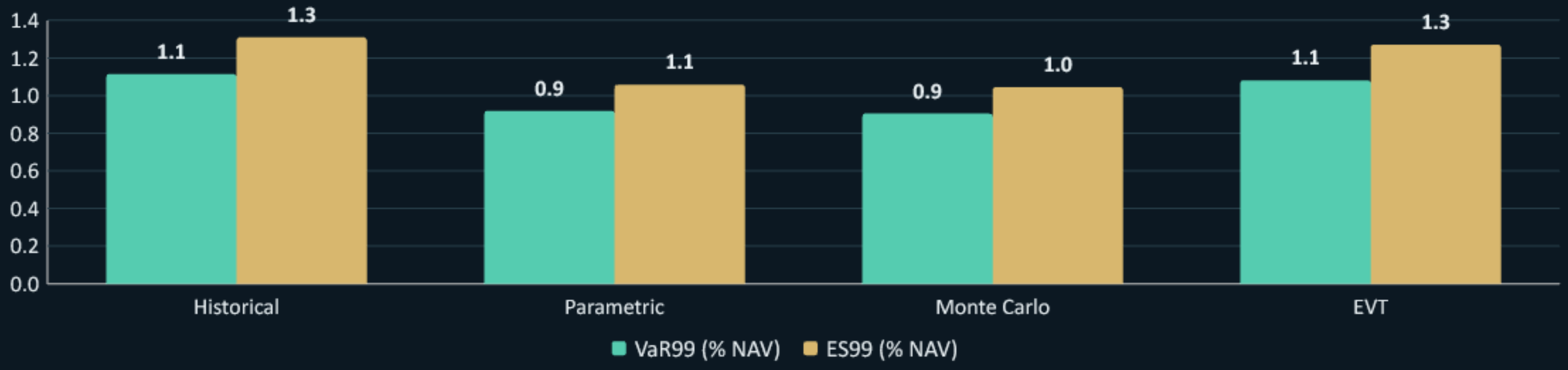
# Return is one part of suitability

AEGIS net held-out CAGR 9.99%
Annual volatility 6.64% · drawdown 6.12%

| Portfolio | Net CAGR | Volatility | Max DD |
|---|---|---|---|
| AEGIS minimum variance | 9.99% | 6.64% | 6.12% |
| Mean variance | 12.78% | 7.81% | 6.49% |
| Constrained risk parity | 11.23% | 7.11% | 5.80% |
| Equal weight | 19.81% | 11.93% | 12.32% |
| 60/40 benchmark | 14.16% | 9.62% | 11.29% |

# Risk estimates depend on assumptions

One-day 99% losses on the same portfolio
Gaussian agreement does not establish tail adequacy



# EVT makes uncertainty visible

90% threshold · 76 exceedances
Shape -0.189; uncertainty spans zero
Tail laboratory exposes threshold sensitivity

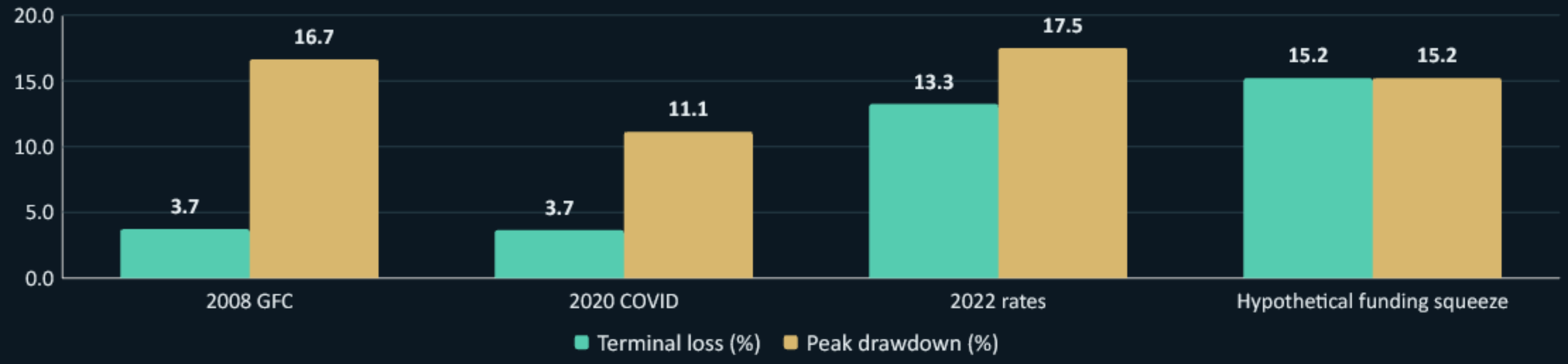
# Forecasts precede observed losses

756 prior observations for every date
Legacy Basel traffic lights use exactly 250 days at 99%

| 99% model | Exceptions | Kupiec p | CC p | Last250 |
|---|---|---|---|---|
| EVT | 6 | 0.234 | 0.474 | Green |
| Historical | 8 | 0.640 | 0.837 | Green |
| Monte Carlo | 13 | 0.263 | 0.445 | Amber |
| Parametric | 9 | 0.898 | 0.909 | Amber |
| Volatility state | 15 | 0.090 | 0.016 | Amber |

# Crisis losses challenge the risk budget

Terminal return can conceal a deeper interim drawdown
Historical replay is hindsight exposure, not a live track record



# Useful innovation includes rejection

Volatility-state score difference +0.132
95% interval [-0.111, +0.394]
No reliable improvement; retain historical baseline

# Conditions for a fundable mandate

Agree crisis tolerance and payout requirement
Validate an eligible allocation and feasible rebalancing
Human team reviews, revises and signs the decision